Figure 8. Layout-to-image synthesis with an *LDM* on COCO [4], see Sec. 4.3.1. Quantitative evaluation in the supplement D.3.

4.3. Conditional Latent Diffusion

4.3.1 Transformer Encoders for LDMs

By introducing cross-attention based conditioning into LDMs we open them up for various conditioning modalities previously unexplored for diffusion models. For **text-to-image** image modeling, we train a 1.45B parameter *KL*-regularized *LDM* conditioned on language prompts on LAION-400M [78]. We employ the BERT-tokenizer [14] and implement τ_θ as a transformer [97] to infer a latent code which is mapped into the UNet via (multi-head) cross-attention (Sec. 3.3). This combination of domain specific experts for learning a language representation and visual synthesis results in a powerful model, which generalizes well to complex, user-defined text prompts, *cf*. Fig. 8 and 5. For quantitative analysis, we follow prior work and evaluate text-to-image generation on the MS-COCO [51] validation set, where our model improves upon powerful AR [17, 66] and GAN-based [109] methods, *cf*. Tab. 2. We note that applying classifier-free diffusion guidance [32] greatly boosts sample quality, such that the guided *LDM-KL-8-G* is on par with the recent state-of-the-art AR [26] and diffusion models [59] for text-to-image synthesis, while substantially reducing parameter count. To further analyze the flexibility of the cross-attention based conditioning mechanism we also train models to synthesize images based on **semantic layouts** on OpenImages [49], and finetune on COCO [4], see Fig. 8. See Sec. D.3 for the quantitative evaluation and implementation details.

Lastly, following prior work [3, 15, 21, 23], we evaluate our best-performing **class-conditional** ImageNet models with $f \in \{4, 8\}$ from Sec. 4.1 in Tab. 3, Fig. 4 and Sec. D.4. Here we outperform the state of the art diffusion model ADM [15] while significantly reducing computational requirements and parameter count, *cf*. Tab 18.

4.3.2 Convolutional Sampling Beyond 256^2

By concatenating spatially aligned conditioning information to the input of ϵ_θ, *LDMs* can serve as efficient general-purpose image-to-image translation models. We use this to train models for semantic synthesis, super-resolution (Sec. 4.4) and inpainting (Sec. 4.5). For semantic synthesis, we use images of landscapes paired with semantic maps [23, 61] and concatenate downsampled versions of the semantic maps with the latent image representation of a $f = 4$ model (VQ-reg., see Tab. 8). We train on an input resolution of 256^2 (crops from 384^2) but find that our model generalizes to larger resolutions and can generate images up to the megapixel regime when evaluated in a convolutional manner (see Fig. 9). We exploit this behavior to also apply the super-resolution models in Sec. 4.4 and the inpainting models in Sec. 4.5 to generate large images between 512^2 and 1024^2. For this application, the signal-to-noise ratio (induced by the scale of the latent space) significantly affects the results. In Sec. D.1 we illustrate this when learning an LDM on (i) the latent space as provided by a $f = 4$ model (KL-reg., see Tab. 8), and (ii) a rescaled version, scaled by the component-wise standard deviation.

The latter, in combination with classifier-free guidance [32], also enables the direct synthesis of $> 256^2$ images for the text-conditional *LDM-KL-8-G* as in Fig. 13.

Method	FID↓	IS↑	Precision↑	Recall↑	N_{params}	
BigGan-deep [3]	6.95	$203.6_{\pm 2.6}$	**0.87**	0.28	340M	-
ADM [15]	10.94	100.98	0.69	**0.63**	554M	250 DDIM steps
ADM-G [15]	4.59	186.7	0.82	0.52	608M	250 DDIM steps
LDM-4 (ours)	10.56	$103.49_{\pm 1.24}$	0.71	0.62	400M	250 DDIM steps
LDM-4-G (ours)	**3.60**	$\mathbf{247.67}_{\pm 5.59}$	**0.87**	0.48	400M	250 steps, c.f.g [32], $s = 1.5$

Table 3. Comparison of a class-conditional ImageNet *LDM* with recent state-of-the-art methods for class-conditional image generation on ImageNet [12]. A more detailed comparison with additional baselines can be found in D.4, Tab. 10 and F. *c.f.g.* denotes classifier-free guidance with a scale s as proposed in [32].

Figure 9. A *LDM* trained on 256^2 resolution can generalize to larger resolution (here: 512×1024) for spatially conditioned tasks such as semantic synthesis of landscape images. See Sec. 4.3.2.

4.4. Super-Resolution with Latent Diffusion

LDMs can be efficiently trained for super-resolution by diretly conditioning on low-resolution images via concatenation (*cf*. Sec. 3.3). In a first experiment, we follow SR3

Figure 10. ImageNet 64→256 super-resolution on ImageNet-Val. *LDM-SR* has advantages at rendering realistic textures but SR3 can synthesize more coherent fine structures. See appendix for additional samples and cropouts. SR3 results from [72].

[72] and fix the image degradation to a bicubic interpolation with $4\times$-downsampling and train on ImageNet following SR3's data processing pipeline. We use the $f = 4$ autoencoding model pretrained on OpenImages (VQ-reg., *cf*. Tab. 8) and concatenate the low-resolution conditioning y and the inputs to the UNet, *i.e.* τ_θ is the identity. Our qualitative and quantitative results (see Fig. 10 and Tab. 5) show competitive performance and LDM-SR outperforms SR3 in FID while SR3 has a better IS. A simple image regression model achieves the highest PSNR and SSIM scores; however these metrics do not align well with human perception [106] and favor blurriness over imperfectly aligned high frequency details [72]. Further, we conduct a user study comparing the pixel-baseline with LDM-SR. We follow SR3 [72] where human subjects were shown a low-res image in between two high-res images and asked for preference. The results in Tab. 4 affirm the good performance of LDM-SR. PSNR and SSIM can be pushed by using a post-hoc guiding mechanism [15] and we implement this *image-based guider* via a perceptual loss, see Sec. D.6.

	SR on ImageNet		Inpainting on Places	
User Study	Pixel-DM ($f1$)	*LDM-4*	LAMA [88]	*LDM-4*
Task 1: Preference vs GT ↑	16.0%	**30.4%**	13.6%	**21.0%**
Task 2: Preference Score ↑	29.4%	**70.6%**	31.9%	**68.1%**

Table 4. Task 1: Subjects were shown ground truth and generated image and asked for preference. Task 2: Subjects had to decide between two generated images. More details in E.3.6

Since the bicubic degradation process does not generalize well to images which do not follow this pre-processing, we also train a generic model, *LDM-BSR*, by using more diverse degradation. The results are shown in Sec. D.6.1.

Method	FID ↓	IS ↑	PSNR ↑	SSIM ↑	N_{params}	[samples/s] (*)
Image Regression [72]	15.2	121.1	**27.9**	**0.801**	625M	N/A
SR3 [72]	5.2	**180.1**	26.4	0.762	625M	N/A
LDM-4 (ours, 100 steps)	2.8†/4.8‡	166.3	$24.4_{\pm 3.8}$	$0.69_{\pm 0.14}$	**169M**	4.62
emphLDM-4 (ours, big, 100 steps)	**2.4**†/**4.3**‡	174.9	$24.7_{\pm 4.1}$	$0.71_{\pm 0.15}$	552M	4.5
LDM-4 (ours, 50 steps, guiding)	4.4†/6.4‡	153.7	$25.8_{\pm 3.7}$	$0.74_{\pm 0.12}$	184M	0.38

Table 5. $\times 4$ upscaling results on ImageNet-Val. (256^2); †: FID features computed on validation split, ‡: FID features computed on train split; *: Assessed on a NVIDIA A100

Model (*reg.*-type)	train throughput samples/sec.	sampling throughput† @256	@512	train+val hours/epoch	FID@2k epoch 6
LDM-1 (no first stage)	0.11	0.26	0.07	20.66	24.74
LDM-4 (*KL*, w/ attn)	0.32	0.97	0.34	7.66	15.21
LDM-4 (*VQ*, w/ attn)	0.33	0.97	0.34	7.04	14.99
LDM-4 (*VQ*, w/o attn)	0.35	0.99	0.36	6.66	15.95

Table 6. Assessing inpainting efficiency. †: Deviations from Fig. 7 due to varying GPU settings/batch sizes *cf*. the supplement.

4.5. Inpainting with Latent Diffusion

Inpainting is the task of filling masked regions of an image with new content either because parts of the image are are corrupted or to replace existing but undesired content within the image. We evaluate how our general approach for conditional image generation compares to more specialized, state-of-the-art approaches for this task. Our evaluation follows the protocol of LaMa [88], a recent inpainting model that introduces a specialized architecture relying on Fast Fourier Convolutions [8]. The exact training & evaluation protocol on Places [108] is described in Sec. E.2.2.

We first analyze the effect of different design choices for the first stage. In particular, we compare the inpainting efficiency of *LDM-1* (*i.e.* a pixel-based conditional DM) with *LDM-4*, for both *KL* and *VQ* regularizations, as well as *VQ-LDM-4* without any attention in the first stage (see Tab. 8), where the latter reduces GPU memory for decoding at high resolutions. For comparability, we fix the number of parameters for all models. Tab. 6 reports the training and sampling throughput at resolution 256^2 and 512^2, the total training time in hours per epoch and the FID score on the validation split after six epochs. Overall, we observe a speed-up of at least $2.7\times$ between pixel- and latent-based diffusion models while improving FID scores by a factor of at least $1.6\times$.

The comparison with other inpainting approaches in Tab. 7 shows that our model with attention improves the overall image quality as measured by FID over that of [88]. LPIPS between the unmasked images and our samples is slightly higher than that of [88]. We attribute this to [88] only producing a single result which tends to recover more of an average image compared to the diverse results produced by our LDM *cf*. Fig. 21. Additionally in a user study (Tab. 4) human subjects favor our results over those of [88].

Based on these initial results, we also trained a larger diffusion model (*big* in Tab. 7) in the latent space of the *VQ*-regularized first stage without attention. Following [15], the UNet of this diffusion model uses attention layers on three levels of its feature hierarchy, the BigGAN [3] residual block for up- and downsampling and has 387M parameters

Figure 11. Qualitative results on object removal with our *big, w/ ft* inpainting model. For more results, see Fig. 22.

instead of 215M. After training, we noticed a discrepancy in the quality of samples produced at resolutions 256^2 and 512^2, which we hypothesize to be caused by the additional attention modules. However, fine-tuning the model for half an epoch at resolution 512^2 allows the model to adjust to the new feature statistics and sets a new state of the art FID on image inpainting (*big, w/o attn, w/ ft* in Tab. 7, Fig. 11.).

5. Limitations & Societal Impact

Limitations While LDMs significantly reduce computational requirements compared to pixel-based approaches, their sequential sampling process is still slower than that of GANs. Moreover, the use of LDMs can be questionable when high precision is required: although the loss of image quality is very small in our $f = 4$ autoencoding models (see Fig. 1), their reconstruction capability can become a bottleneck for tasks that require fine-grained accuracy in pixel space. We assume that our superresolution models (Sec. 4.4) are already somewhat limited in this respect.

Societal Impact Generative models for media like imagery are a double-edged sword: On the one hand, they enable various creative applications, and in particular approaches like ours that reduce the cost of training and inference have the potential to facilitate access to this technology and democratize its exploration. On the other hand, it also means that it becomes easier to create and disseminate manipulated data or spread misinformation and spam. In particular, the deliberate manipulation of images ("deep fakes") is a common problem in this context, and women in particular are disproportionately affected by it [13, 24].

Generative models can also reveal their training data [5, 90], which is of great concern when the data contain sensitive or personal information and were collected without explicit consent. However, the extent to which this also applies to DMs of images is not yet fully understood.

Finally, deep learning modules tend to reproduce or exacerbate biases that are already present in the data [22, 38, 91]. While diffusion models achieve better coverage of the data distribution than *e.g.* GAN-based approaches, the extent to which our two-stage approach that combines adversarial training and a likelihood-based objective misrepresents the data remains an important research question.

For a more general, detailed discussion of the ethical considerations of deep generative models, see *e.g.* [13].

	40-50% masked		All samples	
Method	FID ↓	LPIPS ↓	FID ↓	LPIPS ↓
LDM-4 (ours, big, w/ ft)	**9.39**	0.246± 0.042	**1.50**	0.137± 0.080
LDM-4 (ours, big, w/o ft)	12.89	0.257± 0.047	2.40	0.142± 0.085
LDM-4 (ours, w/ attn)	11.87	0.257± 0.042	2.15	0.144± 0.084
LDM-4 (ours, w/o attn)	12.60	0.259± 0.041	2.37	0.145± 0.084
LaMa [88]†	12.31	**0.243**± 0.038	2.23	**0.134**± 0.080
LaMa [88]	12.0	**0.24**	2.21	0.14
CoModGAN [107]	10.4	0.26	1.82	0.15
RegionWise [52]	21.3	0.27	4.75	0.15
DeepFill v2 [104]	22.1	0.28	5.20	0.16
EdgeConnect [58]	30.5	0.28	8.37	0.16

Table 7. Comparison of inpainting performance on 30k crops of size 512×512 from test images of Places [108]. The column *40-50%* reports metrics computed over hard examples where 40-50% of the image region have to be inpainted. †recomputed on our test set, since the original test set used in [88] was not available.

6. Conclusion

We have presented latent diffusion models, a simple and efficient way to significantly improve both the training and sampling efficiency of denoising diffusion models without degrading their quality. Based on this and our cross-attention conditioning mechanism, our experiments could demonstrate favorable results compared to state-of-the-art methods across a wide range of conditional image synthesis tasks without task-specific architectures.

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